

Homework 5

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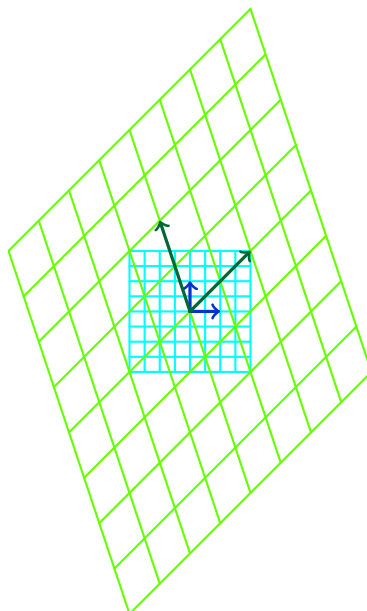
*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

1 Problems from Artin

Problem 1 (I.2.6). Sketch the effect on the plane $\mathbb{R}^2$ of multiplication by the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 3 \end{bmatrix}.$$

Solution.



Problem 2 (I.2.10). Let A be a square matrix. Prove that there exist elementary matrices $E_1, \dots, E_k$

such that

$$E_k \cdots E_1 A$$

is either the identity matrix or has its bottom row equal to zero.

Proof. If A is invertible, then one can transform A to I by elementary row transformation, thus there exists elementary matrices $E_1, \dots, E_k$ such that $E_k \cdots E_1 A = I$.

If A is not invertible, then by elementary row transformation, we can transform A to a row-echelon form, which has its bottom row equal to zero. Thus there exists elementary matrices $E_1, \dots, E_k$ such that $E_k \cdots E_1 A$ has its bottom row equal to zero. $\square$

Problem 3 (I.4.3). Prove that a matrix in which each row contains exactly one entry equal to 1, each column also contains exactly one entry equal to 1, and all other entries are 0, is a permutation matrix.

Proof. Denote the matrix by A . By the definition of A , we can write

$$A = \sum_{i=1}^n E_{i, \sigma(i)}, \quad \sigma(i) \in \{1, 2, \dots, n\}.$$

Moreover, since each column of A contains exactly one entry equal to 1, we have $\sigma(i) \neq \sigma(j)$ for $i \neq j$. Thus σ is a permutation of $\{1, 2, \dots, n\}$, i.e. $\sigma \in S_n$. Therefore A is a permutation matrix. $\square$

Problem 4 (I.4.6). What is the permutation matrix corresponding to the permutation $i \mapsto n - i$?

Solution.

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Problem 5 (I.5.3). Let $A = (a_{ij})$ be an $n \times n$ matrix with integer entries. Prove that A^{-1} has integer entries if and only if

$$\det(A) = \pm 1.$$

Proof. If A^{-1} has integer entries, then $\det A^{-1} \in \mathbb{Z}$. Since $\det(A^{-1}) = (\det A)^{-1}$, we have $(\det A)^{-1} \in \mathbb{Z}$, which implies $\det A = \pm 1$.

Conversely, if $\det A = \pm 1$, then $A^{-1} = (\text{adj } A)/(\det A)$ has integer entries since $\text{adj } A$ has integer entries and $\det A = \pm 1$. $\square$

2 Problems from LADR

Problem 6 (4-10). For $p \in \mathcal{P}(\mathbb{R})$, define $Tp : \mathbb{R} \rightarrow \mathbb{R}$ by

$$(Tp)(x) = \begin{cases} \frac{p(x) - p(3)}{x - 3}, & \text{if } x \neq 3, \\ p'(3), & \text{if } x = 3. \end{cases}$$

for each $x \in \mathbb{R}$. Show that $Tp \in \mathcal{P}(\mathbb{R})$ for every polynomial $p \in \mathcal{P}(\mathbb{R})$ and also show that $T : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ is a linear map.

Proof. Since the polynomial $p(x) - p(3)$ vanishes at $x = 3$, it has a root $x = 3$, hence a factor $x - 3$. Write

$$p(x) - p(3) = q(x)(x - 3).$$

Take derivative of the equation above and set $x = 3$, we can get $p'(3) = q(3)$. Then $T(p) = q \in \mathcal{P}(\mathbb{R})$.

Let's prove that T is a linear map.

- for $p, q \in \mathcal{P}(\mathbb{R})$,

$$T(p + q) = \frac{p(x) + q(x) - p(3) - q(3)}{x - 3} = \frac{p(x) - p(3)}{x - 3} + \frac{q(x) - q(3)}{x - 3} = T(p) + T(q).$$

- for $p \in \mathcal{P}(\mathbb{R})$ and $c \in \mathbb{R}$,

$$T(cp) = \frac{cp(x) - cp(3)}{x - 3} = c \frac{p(x) - p(3)}{x - 3} = cT(p).$$

Therefore T is a linear map. □

Problem 7 (4-12). Suppose m is a nonnegative integer and $p \in \mathcal{P}_m(\mathbb{C})$ is such that there are distinct real numbers $x_0, x_1, \dots, x_m$ with $p(x_k) \in \mathbb{R}$ for each $k = 0, 1, \dots, m$. Prove that all coefficients of p are real.

Proof. By the Lagrange interpolation formula, we can write

$$p(x) = \sum_{k=0}^m p(x_k) \prod_{j \neq k} \frac{x - x_j}{x_k - x_j}.$$

Since $p(x_k) \in \mathbb{R}$ for each $k = 0, 1, \dots, m$, the coefficients of p are all real. □

Problem 8 (4-13). Suppose $p \in \mathcal{P}(\mathbf{F})$ with $p \neq 0$. Let

$$U = \{pq : q \in \mathcal{P}(\mathbf{F})\}.$$

1. Show that $\dim(\mathcal{P}(\mathbf{F})/U) = \deg p$.
2. Find a basis of $\mathcal{P}(\mathbf{F})/U$.

Solution. Let $\deg p = m$. We will show that $\{1 + U, x + U, \dots, x^{m-1} + U\}$ is a basis of $\mathcal{P}(\mathbf{F})/U$.

First, we show that $\{1 + U, x + U, \dots, x^{m-1} + U\}$ is linearly independent. Suppose $c_0, c_1, \dots, c_{m-1} \in \mathbf{F}$ such that

$$c_0(1 + U) + c_1(x + U) + \dots + c_{m-1}(x^{m-1} + U) = 0 + U.$$

Then $c_0 + c_1x + \dots + c_{m-1}x^{m-1} \in U$, which implies that there exists $q \in \mathcal{P}(\mathbf{F})$ such that

$$c_0 + c_1x + \dots + c_{m-1}x^{m-1} = pq.$$

Since $\deg p = m$ and $\deg(q) < m$, this is impossible unless $c_0 = c_1 = \dots = c_{m-1} = 0$. Therefore, $\{1 + U, x + U, \dots, x^{m-1} + U\}$ is linearly independent.

Next, we show that $\{1 + U, x + U, \dots, x^{m-1} + U\}$ spans $\mathcal{P}(\mathbf{F})/U$. Let $r \in \mathcal{P}(\mathbf{F})$. By the division algorithm, there exist unique polynomials $q, r' \in \mathcal{P}(\mathbf{F})$ such that

$$r = pq + r', \quad \deg(r') < \deg(p).$$

Then $r + U = pq + r' + U = r' + U$, and since $r' = \sum_{k=0}^{m-1} a_k x^k$ for some $a_k \in \mathbf{F}$, we have

$$r' + U = \sum_{k=0}^{m-1} a_k (x^k + U).$$

Therefore, $\{1 + U, x + U, \dots, x^{m-1} + U\}$ spans $\mathcal{P}(\mathbf{F})/U$. Hence, it is a basis.

Problem 9 (4-14). Suppose $p, q \in \mathcal{P}(\mathbb{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)–(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that

$$rp + sq = 1.$$

(a) Define

$$T : \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) \rightarrow \mathcal{P}_{m+n-1}(\mathbb{C})$$

by

$$T(r, s) = rp + sq.$$

Show that the linear map T is injective.

(b) Show that the linear map T in (a) is surjective.

(c) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $rp + sq = 1$.

Proof. (a) Suppose $T(r, s) = rp + sq = 0$ for some $r \in \mathcal{P}_{n-1}(\mathbb{C})$, $s \in \mathcal{P}_{m-1}(\mathbb{C})$. Then $rp = -sq$. Since p and q have no common zeros, they are coprime in $\mathbb{C}[x]$. Hence p divides s , but $\deg s < \deg p$, so $s = 0$. Similarly q divides r , and $\deg r < \deg q$ implies $r = 0$. Thus T is injective.

(b) Since $\dim(\mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C})) = n + m = \dim(\mathcal{P}_{m+n-1}(\mathbb{C}))$, and T is injective, we can conclude that T is surjective.

(c) Since T is surjective, there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $T(r, s) = rp + sq = 1$.

□

Problem 10 (5A-7). Define $T \in \mathcal{L}(\mathbb{F}^3)$ by $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. Find all eigenvalues and eigenvectors of T .

Solution. The represent matrix of T is

$$A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Solving the characteristic polynomial $\det(\lambda I - A) = 0$, we can get the eigenvalues $\lambda = 0, 0, 5$.

For $\lambda = 0$, the eigenvectors is $(x, 0, 0)$; For $\lambda = 5$, the eigenvectors is $(0, 0, x)$.

Problem 11 (5A-8). Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that if λ is an eigenvalue of P , then $\lambda = 0$ or $\lambda = 1$.

Proof. Choose an nonzero eigenvector $v \in V$ with eigenvalue λ . Then $Pv = \lambda v$. Applying P again, we get $P^2v = P(\lambda v) = \lambda Pv = \lambda^2v$. Since $P^2 = P$, we have $Pv = \lambda^2v$. Therefore, $\lambda v = \lambda^2v$, which implies $\lambda = \lambda^2$. Solving this equation, we get $\lambda = 0$ or $\lambda = 1$. $\square$

Problem 12 (5A-15). Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{F}$. Show that λ is an eigenvalue of T if and only if λ is an eigenvalue of the dual operator $T^* \in \mathcal{L}(V^*)$.

Proof. Denote M as the represent matrix of T w.r.t basis $\mathcal{B}$. Then M^t is the represent matrix of T^* w.r.t the dual basis $\mathcal{B}^*$. Obviously M and M^t have the same eigenvalues. Therefore, λ is an eigenvalue of T if and only if λ is an eigenvalue of T^* . $\square$

Problem 13 (5A-16). Suppose $v_1, \dots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T , then

$$|\lambda| \leq n \max\{|\mathcal{M}(T)_{j,k}| : 1 \leq j, k \leq n\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j , column k of the matrix of T with respect to the basis $v_1, \dots, v_n$.

Proof. Take an nonzero eigenvector v with eigenvalue λ . Write $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$. Denote $M_{ij} = \mathcal{M}(T)_{ij}$. Then

$$\begin{aligned} Tv = \lambda v &= a_1 \sum_{j=1}^n M_{j1}v_j + a_2 \sum_{j=1}^n M_{j2}v_j + \dots + a_n \sum_{j=1}^n M_{jn}v_j. \\ \implies \lambda a_i &= \sum_{j=1}^n a_j M_{ij}. \end{aligned}$$

Using the triangle inequality, we get

$$\begin{aligned}
 |\lambda| \sum_{i=1}^n |a_i| &= \sum_{i=1}^n \left| \sum_{j=1}^n a_j M_{ij} \right| \leq \sum_{i=1}^n \sum_{j=1}^n |a_j| |M_{ij}| \\
 &\leq \max \{ |M_{ij}| : 1 \leq i, j \leq n \} \sum_{i=1}^n \sum_{j=1}^n |a_j| \\
 &\leq n \max \{ |M_{ij}| : 1 \leq i, j \leq n \} \sum_{j=1}^n |a_j|.
 \end{aligned}$$

Since $v \neq 0$, we have $\sum_{i=1}^n |a_i| > 1$, hence the conclusion follows. $\square$

Problem 14 (5A-22). Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors u and w in V such that

$$T(u) = 3w, \quad T(w) = 3u.$$

Prove that 3 or -3 is an eigenvalue of T .

Proof. If $u + w \neq 0$, then $T(u + w) = 3(u + w)$, which indicates 3 is an eigenvalue of T ; Otherwise, $w = -u$, and $T(u) = -3u$, which indicates -3 is an eigenvalue of T . $\square$